

AS05: Analysis of structural loads.

Group assignment

5.1 Analysis of statistical data.

10-min statistics from structural load measurements on V52 turbine have been stored in the file *46400_AS05_Loads_stats_Spring2025.csv*. These statistics are used to determine the mean loads as function of wind speed. The statistical data file has been provided with the assignment and the channel structure in the file is listed in Table 5.1-1.

Information about the V52 turbine instrumentation can be found in the ‘*Instrumentation for loads measurements*’ presentation.

Q5.1.1 Filtering the data and generating plots.

1. Inspect the provided file and filter out abnormal data from your loads database. e.g. Are your meteorological signals within the range?
2. For the power production MLC, the mode of operation of the wind turbine is ‘running and connected to the grid’. For this specific case, we assumed a filter condition for the rotor speed ($\text{min}(\text{ROT}) > 16.0 \text{ rpm}$) order to ensure the turbine is grid connected. This has already been applied to the raw dataset. Assess operational signals from the turbine and ensure normal operation in your filtered dataset.
3. The wind vane data between 2017-03-30 11:10 & 2017-08-08 13:50, is found to be invalid. Explain why the data from this period needs to be filtered out.
4. As valid measurement sector, consider the reduced $[260^\circ; 320^\circ]$ sector.
5. For this part of the exercise, assume that loads signals’ are calibrated correctly.
6. State if there is any other necessary filters.
7. Bin your filtered database into wind speed bins of 1 m/s width (0-1 m/s, 1-2 m/s, 2-3 m/s, ...).
8. For each structural load channel (marked with yellow in Table 5.1-1), create a 2x1 subplot figure.
 - a. In the top graph, plot as function of $W_{\text{sp_44m}}$, the load channel’s
 - i. 10-min maximum values, and its bin averages
 - ii. 10-min mean, and its bin averages
 - iii. 10-min minimum, and its bin averages
 - b. In the bottom graph, plot the 10-min standard deviation values and its bin averages.

In total, ten 2x1 subplot figures are expected.

Q5.1.2 Conceptual questions with reference to the plots.

Please answer the following questions using the plots made in the previous part and new ones when necessary.

- a) In edgewise bending moment $MyA1$ plot, several levels (trends) can be observed.
 - a. Do you see the same level in all statistics (Mean/Stdev/Min/Max)? Plot the 10-min range (max-min) as function of wind speed and check if you see a similar trend.

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- b. What could cause this?
- c. Plot the signals as function of time to check the evolution of signals in time. You can also consider mapping color to time in the MyA1 stat vs wsp plots.
- d. Same calibration was used for MyA1 signal for the whole dataset: Gain = 763.4 kNm/V, Offset=312.8 kNm. However, it seems that the calibration of this signal changed during the campaign period.
 - i. Try to identify the subsets with different calibrations.
 - ii. Which calibration parameter changed? Gain or offset? Can you suggest the correct calibrations for each subset?
- b) How would electrical power (PO) can be used for the verification check of main shaft torque (MzR) signal?
- c) There is large scatter in tower bottom bending moments' plots. How are $MxTB$ and $MyTB$ changing with the wind direction? Please expand your valid wind direction sector if needed. How could we transform $MxTB$ and $MyTB$ into two other signals where they don't show the same level of scatter? Use, in the explanation, your understanding from wind direction dependency and instrumentation for these signals.
- d) Does the rotor tilt-moment (M_{tilt}) correlate with other loads and what is the zero loading (at low wind speed)?
- e) What does the tower top torsion ($MzTT$) correlate with; yaw or tilt moment?

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5.2 Time series analysis and fatigue load determination.

Input:

Ten complete 10 time series have been recorded and stored in .csv files as described in the Table 5.2-1. These time series files can be found in the folder *46400_AS05_Loads_timeseries.zip* provided with the assignment. The description of the channels is given in Table 5.2-2.

As rainflow algorithm, built-in functions such as ‘rainflow’ in Matlab or Python can be used for counting the load ranges and cycles in a time series signal. Detailed information about these functions are found in Ref. [1] and [2].

Please attach your script in an appendix.

Q5.2.0 Prepare a function in your preferred program s.t.

Inputs: a loads signal in time series (s), sampling frequency (f_s), Wöhler exponent (m)

Process: The rainflow count of the signal s (with f_s sampling frequency) will be computed. Based on ranges and cycles outputs from rainflow count, and the equation from the lecture, the 1-Hz Damage Equivalent Value will be calculated.

Output: 1-Hz Damage Equivalent Value (DEL)

This function can be used to compute DEL values asked in the next questions.

Q5.2.1 Calculate the 1-Hz equivalent load values for m=3, 6, 9 & 12 for: *MxA1*, *MyA1*, *MxR*, *MyR*, *MzR*, *Mtilt*, *Myaw*, *MzTT*, *MyTB* & *MzTB* signals in each of the 10 time series provided as input. Present only your results for the time series recorded at **21/3-2017; 14:30** in a (4 x 10) table.

Q5.2.2 In the statistical data file given as input to previous section (5.1), the DEL values for the 10 time series were filled with an error code. Replace these error codes with what you calculated in the previous question.

Q5.2.3 Use the updated filtered database to plot:

- MxA1_DEL12* and *MyA1_DEL12* values as function of wind speed (ws44).
- DEL3 values of the remaining load signals as function of wind speed (ws44).

Group similar signals together instead of making plot for each load signal.

Q5.2.4 Please discuss:

- What trends do you see in the Q5.2.3 plots?
- Why are the (m=12) DEL values are presented for blade bending moments instead of (m=3)?
- Inspect the time series (TB=201703011200). What is happening halfway through time series (t=270 sec) for *MxTB* signal? Check how other load signals, operational or environmental conditions are changing around that time.

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Annex A: Input to assignment

Table 5.1-1: Channel structure in *46400_AS05_Loads_stats_Spring2025.csv* where 10-min statistics from structural load measurements on V52 turbine are stored.

Column no	Channel name	Unit	Description
1	rname	-	Runname = timestamp with reference to the starting time for the 10 minute period. In yyyyMMddHHmm format.
2	Wsp_44m	m/s	Mean wind speed measured at 44m with cup anemometer mounted on South boom.
3	Wdir_41m	deg	Mean wind direction measured at 41m with wind vane mounted on North boom.
4	ROT	rpm	Mean low speed shaft speed.
5	yaw	deg	Mean yaw position.
6	pa	deg	Mean pitch angle.
7	po	kW	Mean active power.
8	wsn	m/s	Mean wind speed measured on nacelle.
9....16	MyTB..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Tower bottom bending – My TB 050- 230 Deg
17....24	MxTB..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Tower bottom bending – Mx TB 150- 328 Deg
25....32	MzTT..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Tower top torsion
33....40	Myaw..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Derived main shaft yaw moment
41....48	Mtilt..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Derived main shaft tilt moment
49....56	MzR..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Shaft torque
57....64	MyR..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Main shaft bending moment – MyR 90-270deg
65....72	MxR..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Main shaft bending moment – MxR 00-180deg
73....80	MyA1..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Edgewise bending moment at the root of blade A
81....88	MxA1..	kNm	Mean/Stdev/Min/Max/DEL3/DEL6/DEL9/DEL12 Flapwise bending moment at the root of blade A

Table 5.2-1: List of time series stored as TB=rname in folder [46400_AS05_Loads_timeseries.zip](#)

status	ws44	ti44	TB = rname
production	3.26	21.49	201804050900
production	4.7	22.39	201703141000
production	6.51	20.63	201703031120
production	8.95	20.17	201703011200
production	9.33	6.13	201703100900
production	10.19	23.56	201806201550
production	12.06	8.52	201704061420
production	13.36	19.07	201703211430
production	15.61	16.06	201702230240
production	18.15	8.97	201701032150

Table 5.2-2- List of channels

cid	name	unit	description
time	time	sec	time signal for the time series
11	ws70	m/s	Wind speed, h=70m (cup anemometer)
13	ws44	m/s	Wind speed, h=44m (cup anemometer)
15	ws18	m/s	Wind speed, h=18m (cup anemometer)
107	wd70	deg	Wind direction, h=70m (3D sonic)
31	wd41	deg	Wind direction, h=41m (Vane)
202	ROT	rpm	low speed shaft
205	pa	deg	Pitch angle
207	po	kW	Active power
210	yaw	Deg	Yaw position
201	wsn	m/s	Wind speed measured on nacelle
211	Azi	deg	Azimuth position of rotor
231	MxA1	kNm	Flapwise bending in blade A root
232	MyA1	kNm	Edgewise bending in blade A root
261	MxR	kNm	Bending shaft moment - MxR 00-180deg
262	MyR	kNm	Bending shaft moment - MyR 90-270deg
263	MzR	kNm	Shaft torque - MzR
264	Mtilt	kNm	Derived tilt moment, Mtilt
265	Myaw	kNm	Derived tilt moment, Myaw
273	MzTT	kNm	Tower top torsion - MzTT
274	MxTB	kNm	Tower bottom bending - MxTB 150 - 328 Deg
275	MyTB	kNm	Tower bottom bending - MyTB 050 - 230 Deg



References

[1]: <https://se.mathworks.com/help/signal/ref/rainflow.html>

[2]: <https://pypi.org/project/rainflow/>

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Group assignment

Remember to fill in the table describing the contribution from the group members.

Questions/weight	Name#1	Name#2	Name#3	Final mark
Q5.1 /60%	%	%	%	100%
Q5.2 /40%	%	%	%	100%
Rating				100%

Required: An assignment report of including a table with student contributions. The report should present the problem(s) and how the problems have been solved.

Furthermore, each question should be addressed clearly and discussed when necessary.

The group assignment should include:

- 1) Name, student number and contribution to each question (matrix above);
- 2) Description of the data processing and analysis;
- 3) Address of each question and their solution;
- 4) Scripts used for the assignment (in Appendix).

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